Availability Numbers (High Availability in System Design)

High Availability (HA) is a key **non-functional requirement** in distributed systems that defines how reliably a system stays operational over time.

1. Core Idea

High availability means a system is **continuously operational with minimal downtime**.

```None
100% availability → 0 downtime (theoretical, not achievable in real systems)
```

Real-world systems are always designed to **tolerate failures, not avoid them completely**.

2. What is Availability?

Availability is measured as the **percentage of uptime over total time**.

```None
Availability = (Uptime / Total Time) × 100
```

- Higher percentage → more reliable system
- Even small downtime becomes critical at scale (millions of users)

3. "Nines" Concept (Very Important)

Availability is commonly expressed in terms of **nines (9s)**.

Availability	Downtime per Year	Downtime per Month	Level
99%	~3.65 days	~7.2 hours	Low
99.9%	~8.76 hours	~43 minutes	Standard (most cloud services)
99.99%	~52 minutes	~4.3 minutes	High Availability
99.999%	~5 minutes	~26 seconds	Ultra High Availability

More “9s” = exponentially harder to achieve reliability.

4. SLA (Service Level Agreement)

A **Service Level Agreement (SLA)** is a formal contract between:

- **Service Provider** (AWS, Google Cloud, Microsoft, or your system)
- **Customer**

It defines:

- Expected **uptime guarantee**
- Allowed **downtime**
- Penalties or compensation if violated

```None
SLA = Formal uptime guarantee + penalty agreement
```

Real-World Examples

Cloud providers typically guarantee:

```None
AWS / Google Cloud / Azure → 99.9% or higher SLA
```

Higher tiers (99.99%+) are used for critical systems like payments and banking.

5. SLA vs SLO vs SLI (Important Interview Concept)

Term	Meaning
SLI	Measured metric (actual uptime, latency, error rate)
SLO	Internal reliability target (engineering goal)
SLA	External contractual guarantee to customers

Interview tip: Mention all three for strong system design answers.

6. Why Availability Matters

High availability ensures:

- System remains **reliable under failures**
- Better **user experience**
- Reduced **business revenue loss**
- Higher **trust and scalability**

7. How High Availability is Achieved

To achieve high availability, systems use:

✔ Redundancy

Multiple servers to avoid single point of failure

✔ Load Balancing

Distributes traffic across multiple instances

✔ Failover Systems

Automatically switches to backup system on failure

✔ Replication

Data copied across multiple nodes or regions

✔ Health Checks

Detect and remove unhealthy instances

✔ Auto Scaling

Dynamically handles traffic spikes

8. Real-World Insight (Very Important)

- 100% uptime is impossible in distributed systems
- Even major companies (Google, Amazon, Meta) experience outages
- System design goal is:

None
```
Design for failure, not avoidance of failure
```

9. Mental Model for Interviews

Think of availability as:

None
```
More redundancy + faster recovery = higher availability
```

Or simply:

None
```
Availability = How quickly system recovers from failures + how
well it avoids single points of failure
```

10. One-Line Summary

Availability is the percentage of system uptime, commonly measured in “nines”, and achieved through redundancy, replication, failover mechanisms, and fault-tolerant system design.